System Design Document

AI-Powered Education Student Success Assistant · End-to-End GenAI Architecture

1. Problem Statement

Higher education distance-learning institutions face persistent student attrition, with over 30% of enrolled learners disengaging or withdrawing before completion. In the Open University Learning Analytics Dataset (OULAD), students generate millions of virtual learning environment (VLE) interaction logs, assessment submissions, and registration events. Academic advising teams cannot manually review these massive data streams in time to conduct early interventions.

Core Questions Addressed:

• *Similarity & Precedents:* Has a student with similar academic and learning-activity patterns previously struggled or withdrawn from a course?

• *Factor Attribution:* What specific behavioural, submission, and engagement factors drive withdrawal risk?

• *Prioritization under Capacity:* Which students should advisors contact first when advisor bandwidth is strictly limited to 10%–20% of the cohort?

Pervasive Target Leakage Resolution: A critical flaw identified in prior learning analytics systems was the inclusion of students who had already unregistered prior to course start or cutoff day. Training on these students yields deceptively high ROC-AUC (>0.90) because absence of clicks trivializes prediction. Our design enforces *temporal in-scope isolation*: only students actively registered at the checkpoint are evaluated.

2. Architecture Design & Pipeline Stages

The architecture strictly satisfies the pipeline flow mandated in the Capstone guidelines: **Dataset** → **Preprocessing** → **Chunking** → **Embeddings** → **Vector Database** → **Retriever** → **LLM / RAG Pipeline** → **Agents** → **API Layer** → **User Interface**.

Pipeline Stage	Technical Implementation	Design Responsibility
1. Dataset	OULAD (7 Relational CSVs, 32.5k records)	Source of ground-truth demographics, submissions, VLE logs
2. Preprocessing	Temporal Cutoffs (Weeks 6, 9, 15, 22)	In-scope filtering; median imputation; sensitive feature isolation
3. Chunking	Discretized Behavioural Tokenizer	Bands numeric metrics into semantic tokens (e.g. score_low)
4. Embeddings	Attribution-Weighted Vectors & Dense	TreeSHAP-weighted Euclidean vectors + qualitative narratives
5. Vector DB	Supervised kNN + BM25 Inverted Index	In-memory indexed historical precedent storage (earlier cohorts)
6. Retriever	Hybrid RRF (Reciprocal Rank Fusion)	Fuses keyword and behavioural kNN; metadata filtering
7. LLM / RAG	Evidence Grounding & Code Guardrails	Numeric grounding validator; anti-punitive & anti-stigma screening
8. Agents	Multi-Agent Workflow (5 roles)	Performance → Risk → Router → Intervention ■ Critic
9. API Layer	FastAPI Microservice (REST + Jobs)	Token-bucket rate limiter; API key auth; Pydantic validation

10. UI Layer	Streamlit Dashboard + React Console	Interactive course clock; student diagnosis; copilot chat

3. Technology Selection & Justifications

• **Machine Learning Engines:** Scikit-Learn (HistGradientBoosting) and XGBoost. Gradient boosted trees provide native missing value handling (critical for assessments not yet due), handle extreme feature scale differences without distortion, and enable exact TreeSHAP attributions.
• **Calibration Strategy:** Raw probabilities from boosted trees are uncalibrated. We apply Leave-One-Presentation-Out Platt scaling to margin outputs, lowering expected calibration error (ECE) to 0.006.
• **Vector Database & Retrieval:** In-memory Supervised kNN index combined with Rank-BM25. Because historical student cases number ~25,000 per module, exact kNN search with feature weighting runs in <10ms without the operational failure surface or latency of external vector microservices.
• **Generative LLM Selection:** Any OpenAI-compatible model (GPT-4o/mini, Claude, or local Ollama). The architecture separates computation from narration: all facts and numbers originate in deterministic code; the LLM merely synthesizes language.

4. Retrieval Strategy & Hybrid Search

The Tabular RAG Dilemma: In numerical learning records, raw figures like 'score: 45' and 'score: 46' share no token overlap in BM25 and cannot be matched semantically by dense text encoders. Our retrieval strategy employs a dual-pronged approach:
1. *Qualitative Discretization (Lexical Channel):* Continuous variables are discretized into cohort-relative semantic bands (e.g., score_failing, scores_declining, engagement_very_low, missed_most). BM25 searches over these discretized tokens.
2. *Supervised Attribution-Weighted kNN (Behavioural Channel):* Robust-scaled feature vectors are multiplied by TreeSHAP mean absolute attributions. Features that drive dropout are weighted proportionally higher in distance calculation.
3. *Reciprocal Rank Fusion (RRF):* Combines rankings with weights tuned on validation presentations.
4. *Temporal Guard:* Students can ONLY be matched against precedents from strictly EARLIER course presentations, preventing future data contamination.

5. Agent Design & Workflow

The multi-agent system orchestrates 5 distinct roles connected via typed Pydantic shared state:
• **Performance Analysis Agent:** Analyzes assessment trajectory, submission deadlines, early/late ratios, and weekly engagement slopes.
• **Risk Classification Agent:** Ingests calibrated probabilities, TreeSHAP attributions, and peer quintiles to assign High/Medium/Low tiers.
• **Routing Agent:** Routes Low-risk students to automated Monitoring; escalates Medium/High risk students to Intervention planning.
• **Intervention Recommendation Agent:** Proposes 2–4 targeted interventions chosen strictly from an approved institutional catalog.
• **Critic Agent & Code Guardrails:** Automatically validates the draft plan against 5 deterministic rules: (1) no ungrounded numbers, (2) no punitive terms, (3) no clinical diagnosis, (4) no sensitive demographic mentions, (5) verified catalog action IDs. If violations occur, the plan is revised once; persistent failures trigger deterministic fallback.

6. Evaluation Strategy & Metrics

Evaluation spans three rigorous tiers:
1. *Predictive Evaluation:* Temporal test on untouched 2014J presentation evaluating ROC-AUC (with 95% bootstrap CI), PR-AUC, Brier score, ECE calibration error, and Precision@Capacity (10%, 20%, 30%).
2. *Retrieval Evaluation:* Benchmarked against random selection using Neighbour-Vote AUC (0.835 vs 0.512) and nDCG@10 (0.866 vs 0.544).
3. *GenAI & LLM-as-Judge Evaluation:* DeepEval and custom LLM judges evaluate faithfulness, non-punitiveness, actionability, and fairness. Deterministic code verifies that 100% of final plans adhere to safety guardrails.

7. Challenges, Limitations & Trade-offs

• **Capacity-Centric Prioritization:** Traditional ML optimizes global F1 or AUC, but academic advisors have finite hours and can only contact 10%–20% of students. We traded global threshold tuning for a simplex grid search that maximizes precision specifically at top-decile capacity.
• **Ethical Privacy Safeguards:** Protected demographic features (gender, IMD deprivation, disability) are completely isolated from model training and prompts. Although sensitive proxies exist in behavioral data, our fairness audit continually monitors equal opportunity gaps across student subgroups.
• **Human-in-the-Loop Governance:** The assistant never autonomously contacts students. Outreach drafts and intervention plans are recommendations requiring explicit advisor review and sign-off.